

Photosynthesis

Introduction to Photosynthesis

Photosynthesis converts light energy into chemical energy stored in glucose.

It happens in the chloroplasts of plant cells.

The overall reaction combines carbon dioxide and water to make sugar.

Photosynthesis

Light Absorption

Chlorophyll absorbs mostly red and blue light.

Green light is reflected, which is why leaves look green.

Carotenoid pigments capture extra wavelengths of light.

Light Reactions

Light energy splits water molecules during photolysis.

Electrons move through the photosystem and release energy.

Photosynthesis

The Calvin Cycle

The Calvin cycle fixes carbon dioxide into organic molecules.

It uses ATP and NADPH produced in the light reactions.

The cycle repeats three times to build one molecule of sugar.